Taxonomy of Black-Box Multiobjective Optimization Algorithms

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October 14, 2025

Multiobjective optimization algorithms that treat the problem objectives and constraints as black boxes can be broadly categorized based on the number of solutions they explore at each iteration. They either focus on a *single solution* or maintain a *population of solutions*. Accordingly, they fall into one of the following two classes:

- **Single-point algorithms**
- **Population-based algorithms**

Regardless of the search strategy, an algorithm can tackle a constrained multiobjective optimization problem (CMOP) in one of two ways. It can solve the original CMOP directly, or it can first transform the CMOP into one or more constrained single-objective optimization problems (CSOPs) and solve those instead. In the latter approach, multiple CSOPs are typically generated and optimized iteratively. Thus, algorithms also fall into one of the following two categories:

- **Algorithms that solve multiple CSOPs**
- **Algorithms that solve the original CMOP**

Note that interactive methods, where a decision maker sets up updates preferences during the search, can be regarded as algorithms that solve multiple CSOPs.

Additionally, population-based algorithms can be further subdivided into algorithms which utilize a single population and those utilizing multiple subpopulations. In the latter, subpopulations are handled in parallel. This leads to a third category:

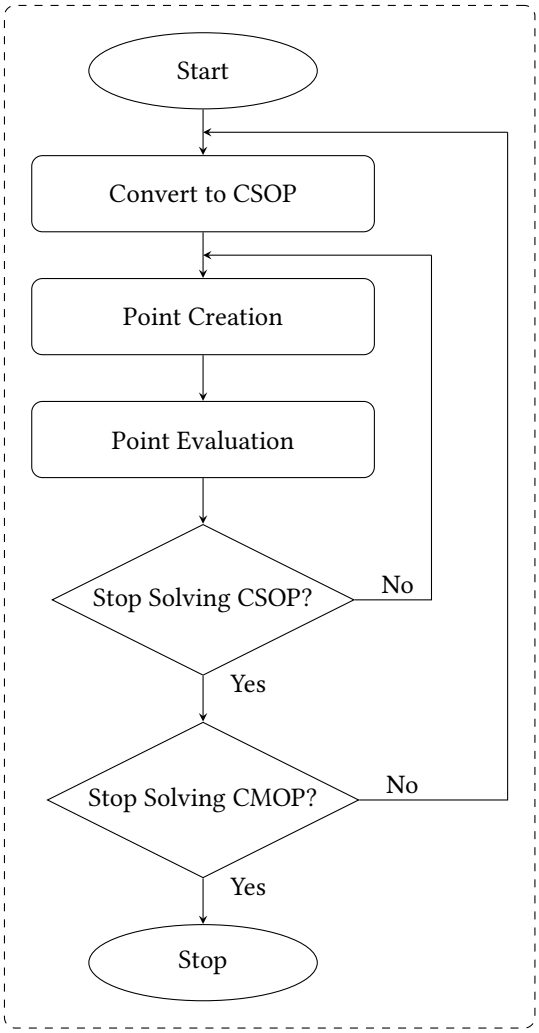
- **Algorithms that solve the original CMOP in parallel (using subpopulations)**

The result is five algorithm categories:

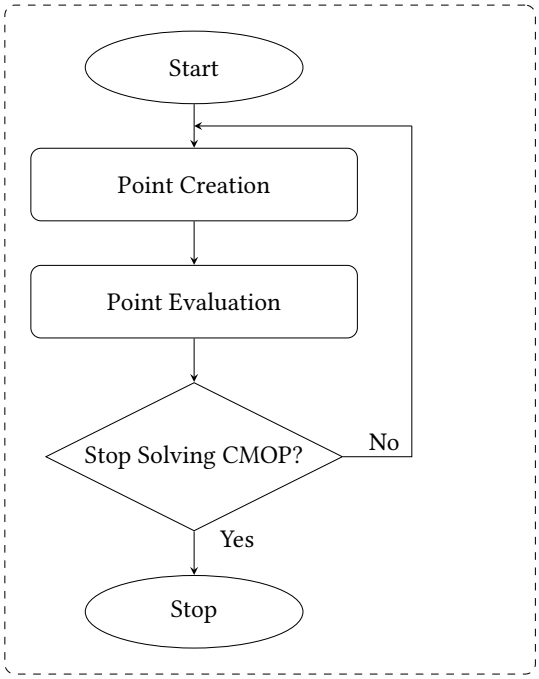
- **Single-point algorithms that solve multiple CSOPs**
- **Single-point algorithms that solve the original CMOP**
- **Population algorithms that solve multiple CSOPs**
- **Population algorithms that solve the original CMOP**
- **Population algorithms that solve the original CMOP in parallel (using subpopulations)**

1 Single-point algorithms

The diagrams below represent single-point algorithms that solve multiple CSOPs and the original CMOP. Examples of the first kind are traditional scalarization methods, such as the weighted sum, the weighted Chebyshev (Tchebycheff), the ε-constraint method etc. A trivial example of the second kind is the random search.



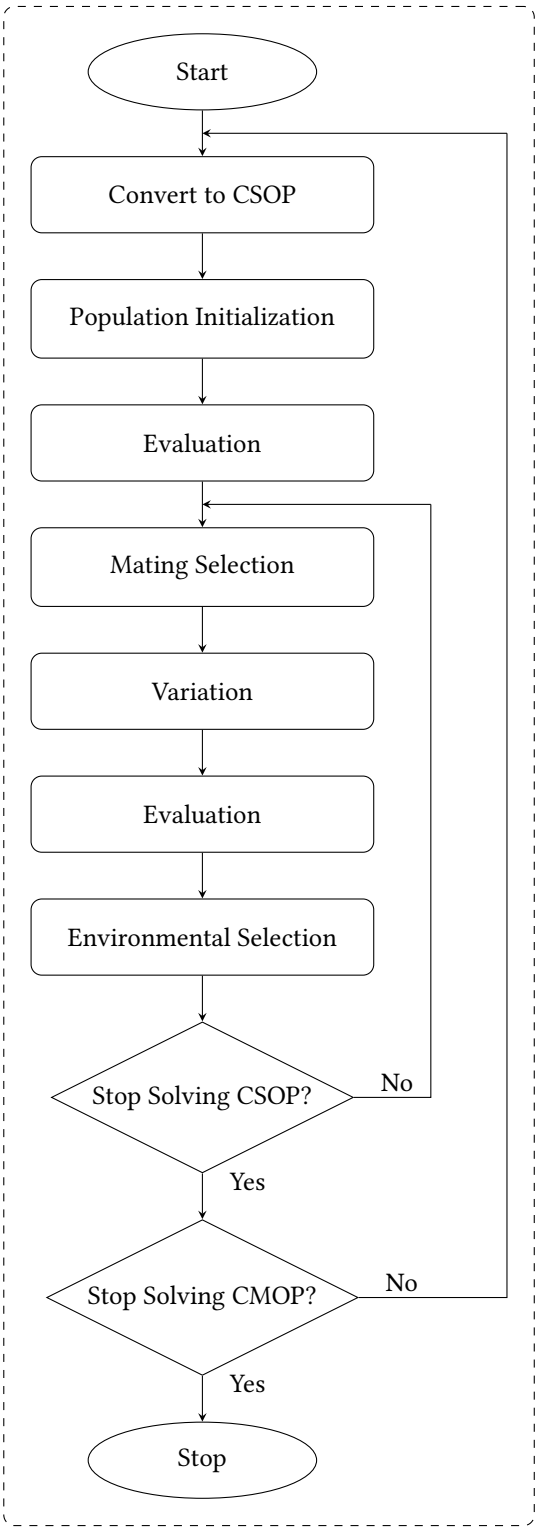
A single-point algorithm that solves multiple CSOPs



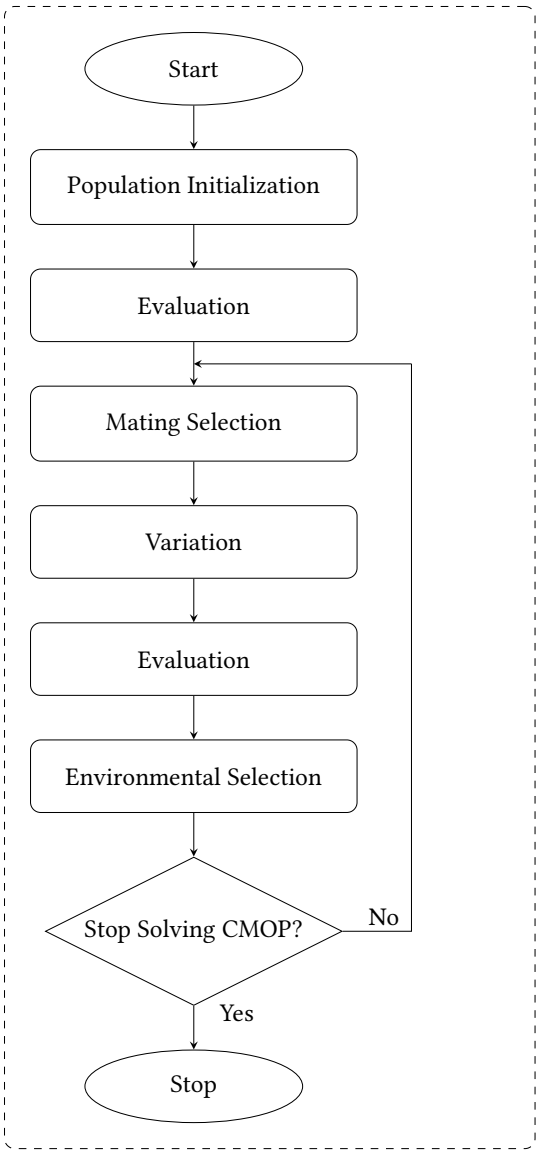
A single-point algorithm that solves the original CMOP

2 Population algorithms

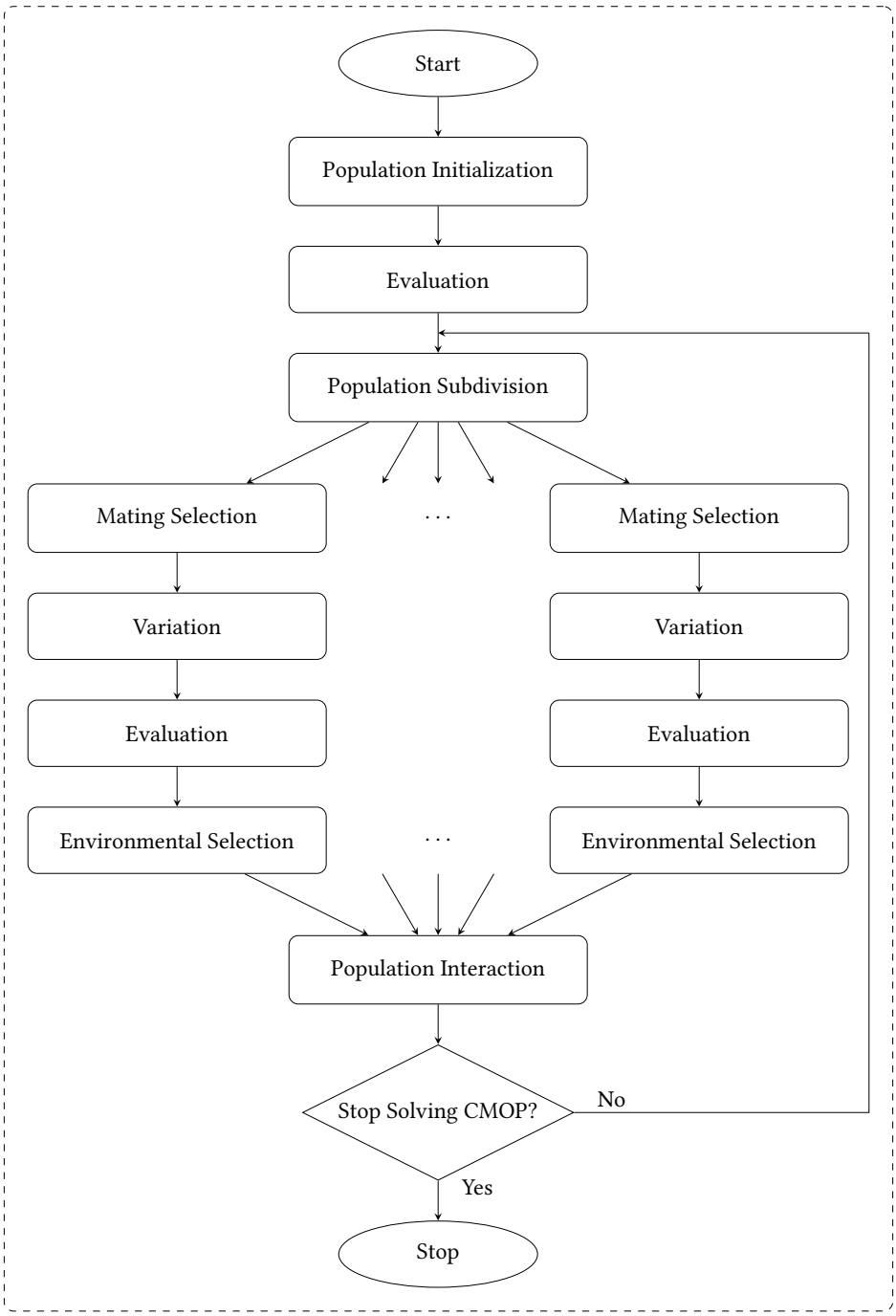
The diagrams below represent population algorithms that solve multiple CSOPs and the original CMOP (both with a single population and multiple subpopulations in parallel). Examples of the first kind are evolutionary algorithms applied to scalarized versions of the CMOP, while examples of the second and third kind are (mostly) multiobjective optimization evolutionary algorithms.



A population algorithm that solves multiple CSOPs



A population algorithm that solves the original CMOP



A population algorithm that solves the original CMOP in parallel (using subpopulations)

3 Algorithm Examples

3.1 Single-point algorithms

Examples of single-point multiobjective optimization algorithms which handle constraints grouped by strategy:

- **Single-point algorithms solving multiple CSOPs**
  - Random search
  - Weighted sum
  - Weighted Chebyshev
  - ε-constraint method [1]
- **Single-point algorithms solving the original CMOP**
  - MO random search [1]
  - DMulti-MADS [2]

3.2 Population algorithms

Examples of population multiobjective optimization algorithms which handle constraints grouped by strategy:

- **Population algorithms solving multiple CSOPs**
  - COMO-CMA-ES [3]
  - Scalarization-based EAs (e.g., Weighted Sum EA, ε-constraint EA) [1]
- **Population algorithms solving the original CMOP**
  - NSGA-II [4]
  - NSGA-III [5]
  - MOEA/DD [6]
  - SPEA2[1] [7]
  - SMS-EMOA[1] [8]
  - PPS [9]
  - GDE3 [10]
- **Population algorithms solving the original CMOP in parallel (using subpopulations)**
  - Island-model parallel NSGA-II [11]
  - CCMO (Coevolutionary constrained multi-objective optimization framework) [12]
  - BiCO (Bidirectional coevolution constrained multiobjective evolutionary algorithm) [13]
  - MCCMO (Multi-population coevolutionary constrained multi-objective optimization) [14]

[1] SPEA2 and SMS-EMOA were not originally proposed with constraint handling mechanisms in their core formulations.

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